Lesson 8:

Creating Templates



IGSS

Interactive Graphical
SCADA System

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Contents of this lesson



Topics:

- What is a template?
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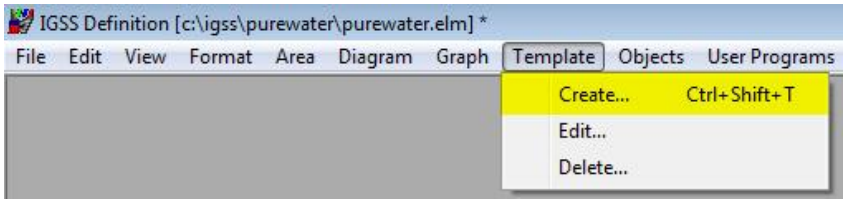
What is a template ?



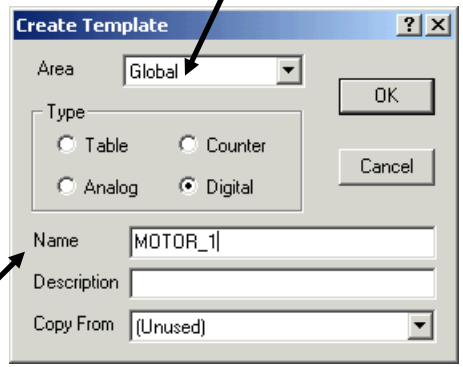
- When a number of process components share several properties, you should use templates
- Analog, table, counter and digital object types are supported
- The real process components (IGSS objects) may be based on templates
- Template-based objects subscribe to changes in the template

Creating templates

Template menu in Definition



Templates defined in the ”Global” area can be used in the entire configuration. Templates defined in other areas are local.



Template name
Recommendation is
ALL CAPS for template names
Normal case for object names

Digital templates

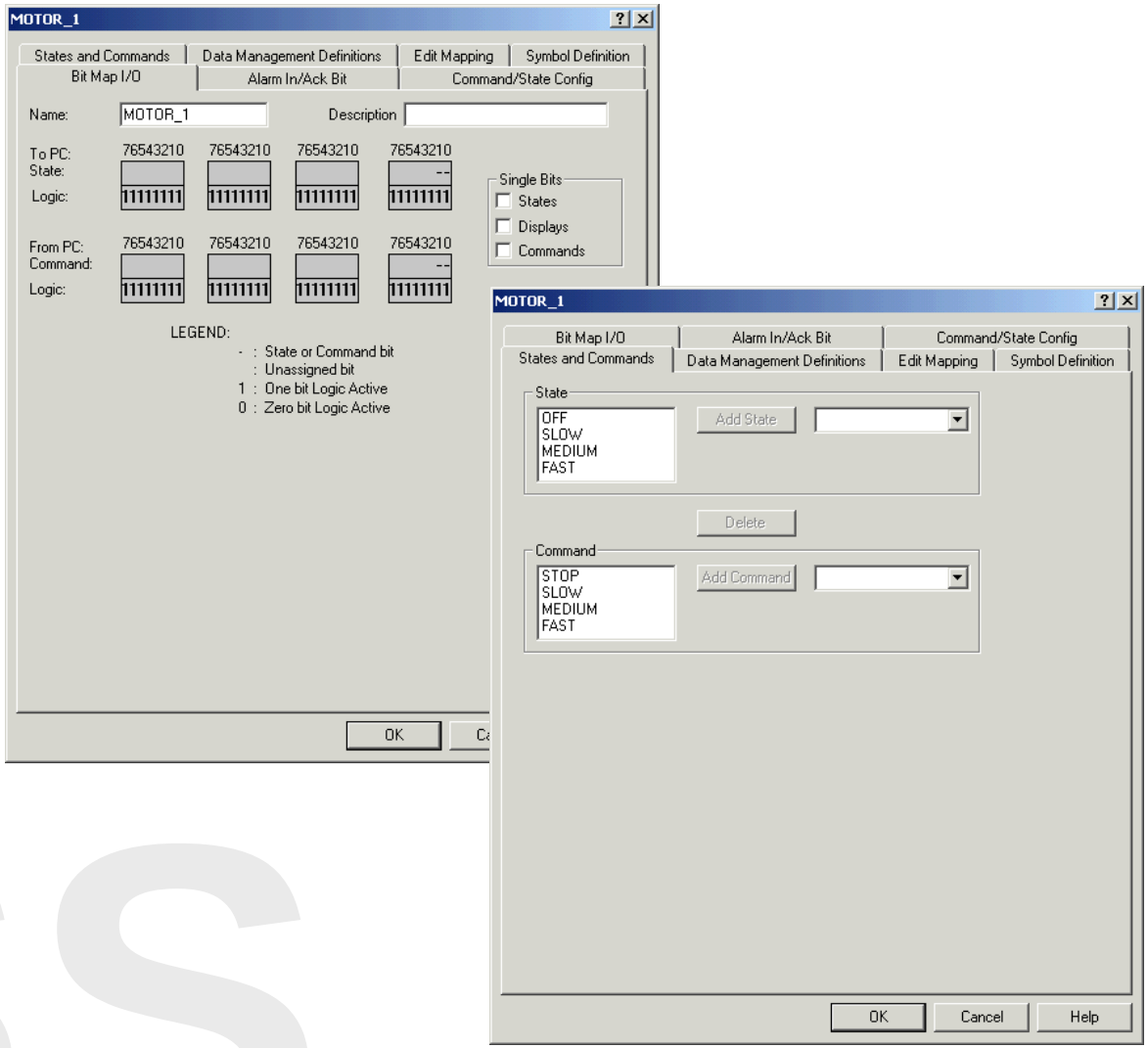


Summary:

- Define states and commands
- Define valid commands
- Define incoming digital alarms
- Define outgoing alarm acknowlegment

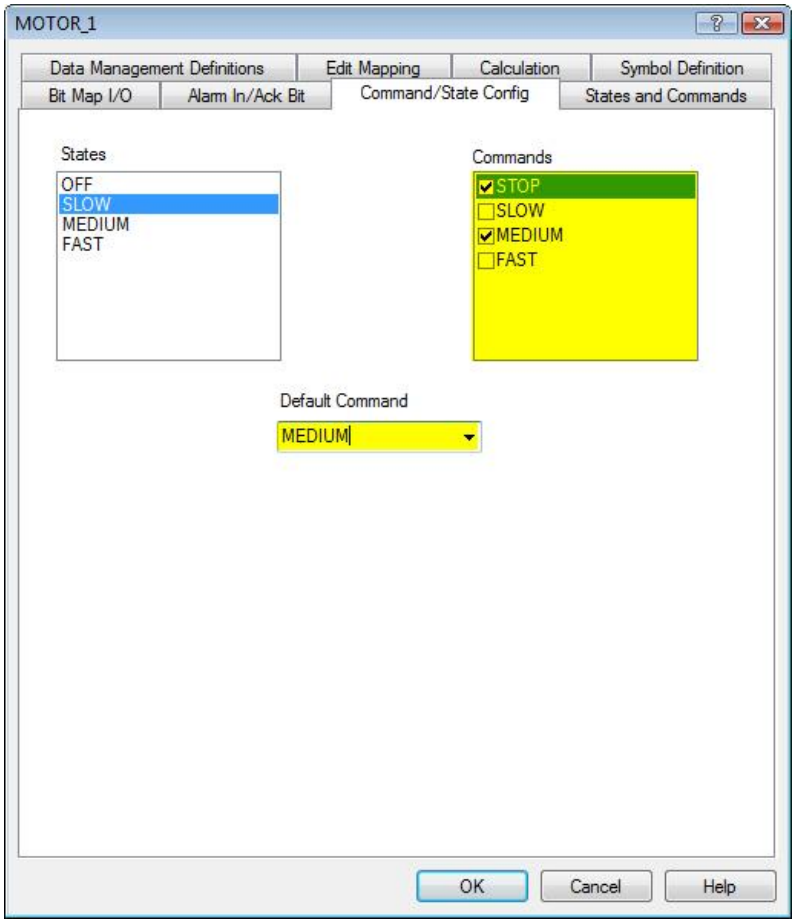
States and commands

1. Define number of states and commands.
2. Assign a name to each state and command.



Define valid commands

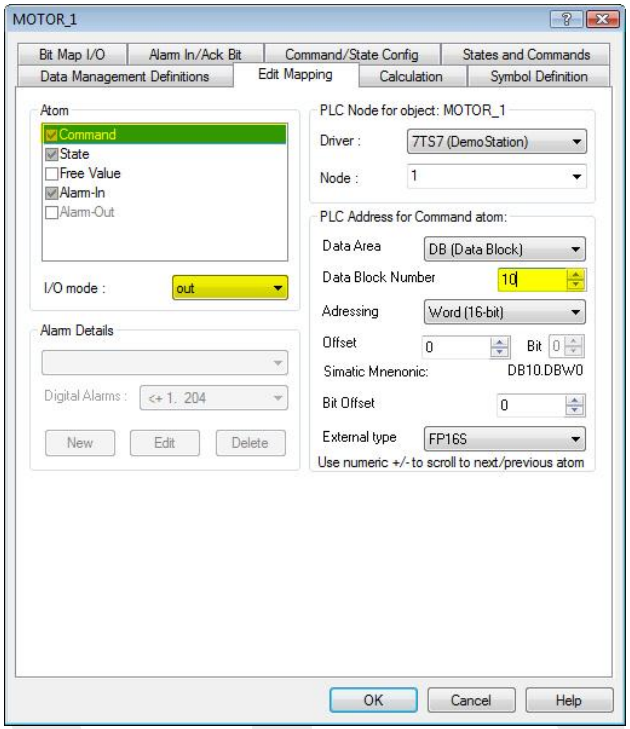
1. Select a state.
2. Select the valid commands for that state.
3. Specify the recommended operator action (command) for the state.

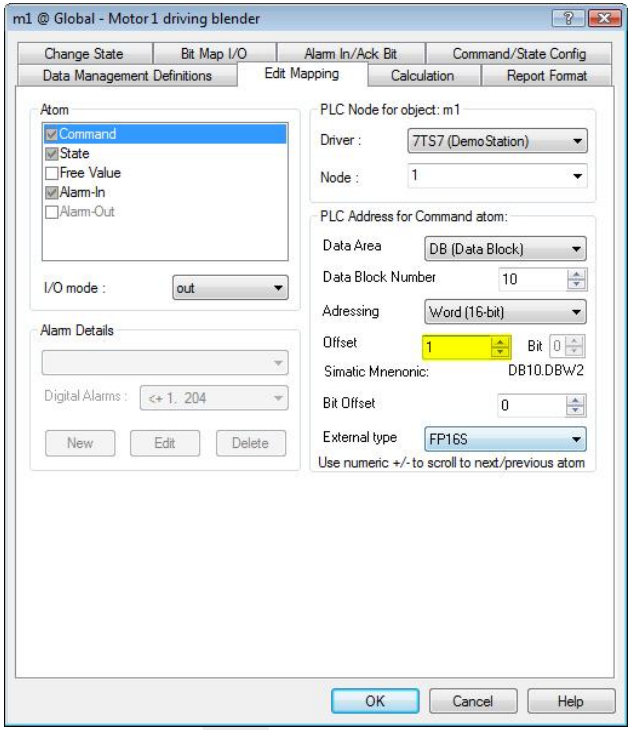


PLC addressing – Commands

Template = MOTOR_1
- Choose atom and I/O mode=”Out”
- Choose driver and node number
- Choose data group and external type
- Leave word offset and bit offset as is

Object = m1, m2, m3, m4
- Type relevant word offset





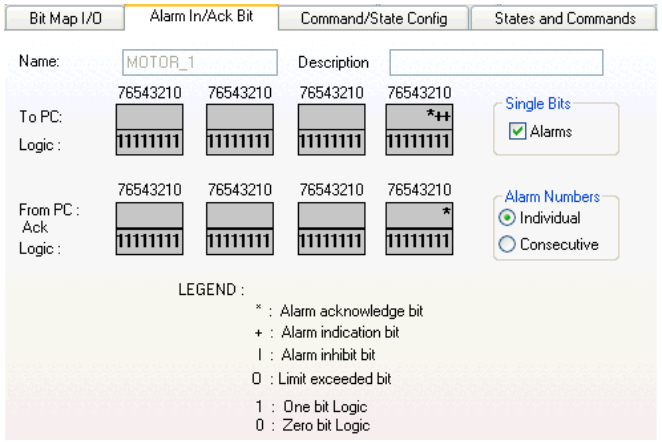
PLC addresses

Obj.	DG	Word	Bit
m1	10	1	0
m2	10	2	0
m3	10	3	0
m4	10	4	0

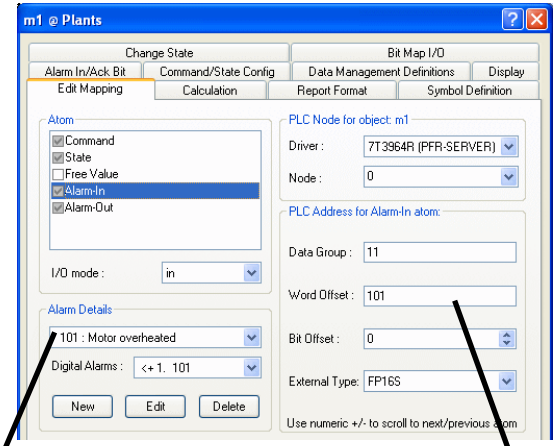
Incoming alarms

Definition

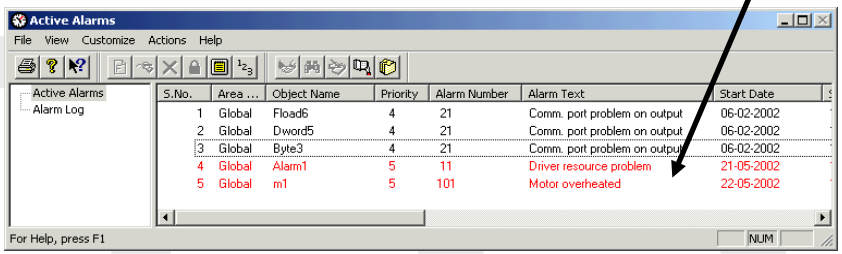
① Two alarm indication bits (+) are defined on the ”Alarm In/Ack Bit” tab.



② The PLC address is specified as the ”Alarm-In” atom on the ”Edit Mapping” tab.



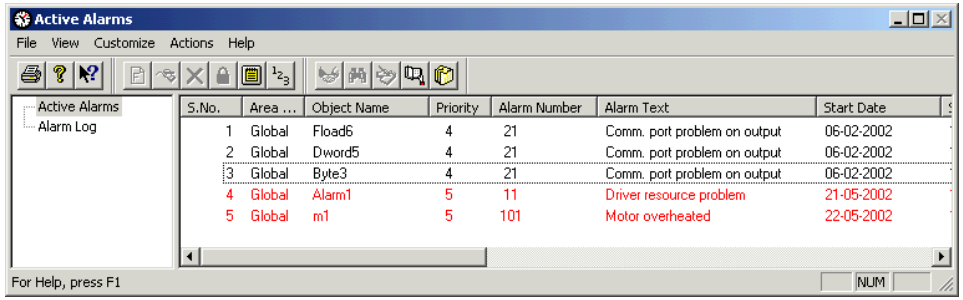
③ When one of these bits is raised, the corresponding alarm text occurs in the Alarm List.



PLC **Data Group 11**

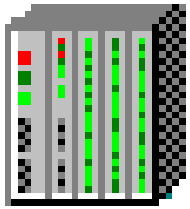
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Word 0																
Word 1																
Word 2																
Word 3																
Word ...																
Word ...																
Word ...																
Word ...																
Word ...																
Word ...																
Word 99																
Word 100																
Word 101															x	x

Alarm acknowledgement



①

The operator acknowledges the alarm. The bit is sent to the PLC address specified for ”Alarm-Out” on the ”Edit Mapping” tab.

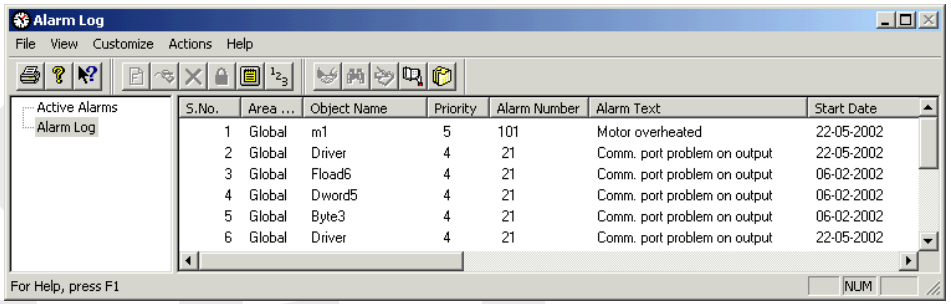


②

The PLC program registers the operator acknowledgement and returns a confirmation bit to IGSS. This the bit specified as * in the ”To PC” field on the ”Alarm In/Ack Bit” tab.
The PLC address is specified is the ”Alarm-In” atom on the ”Edit Mapping” tab.

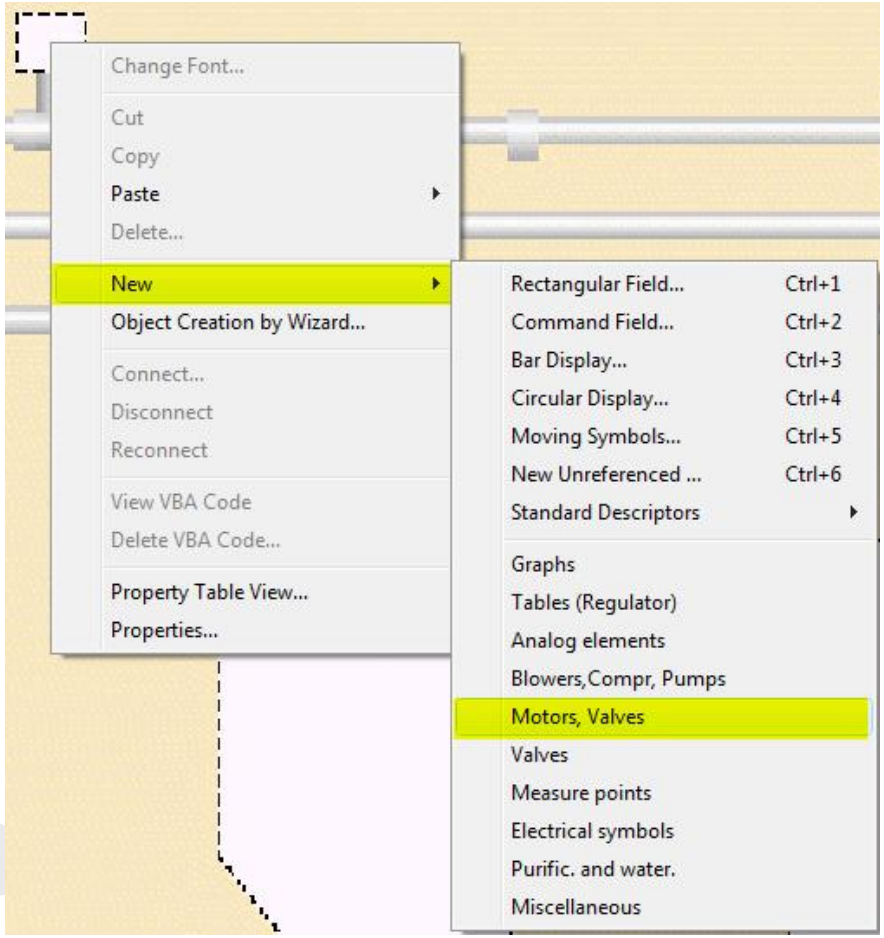
③

When IGSS receives the PLC’s acknowledgement, the alarm is acknowledged in IGSS. It typically changes color in the Alarm List and is moved to the ”Alarm Log”.



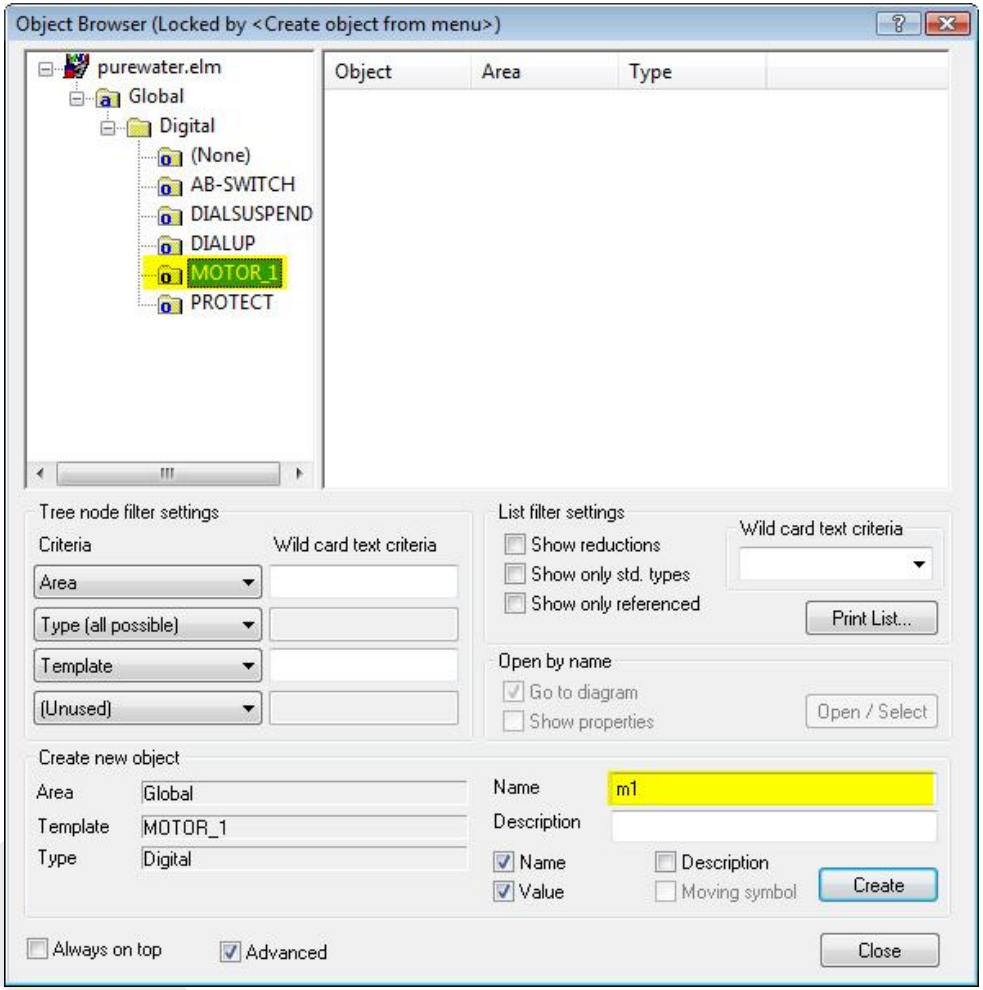
Create object from template

1. Right-click on the diagram.
2. Create a new digital object.



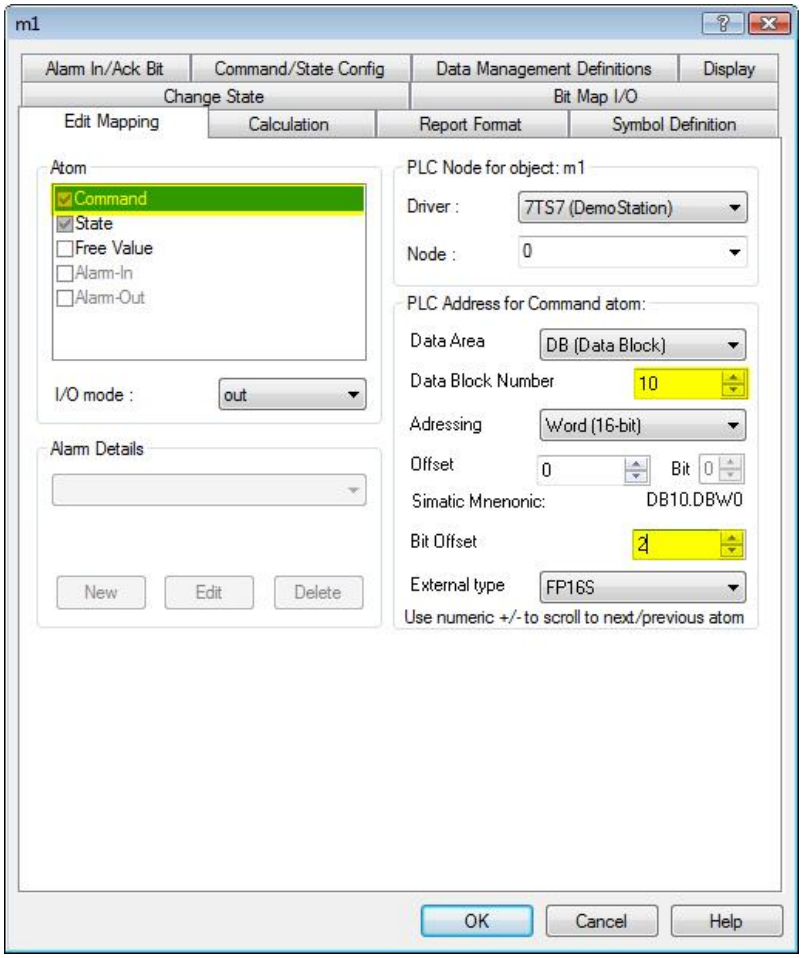
Connect object to template

1. Choose area.
2. Choose template.
3. Type object name.
4. Click **Create**.



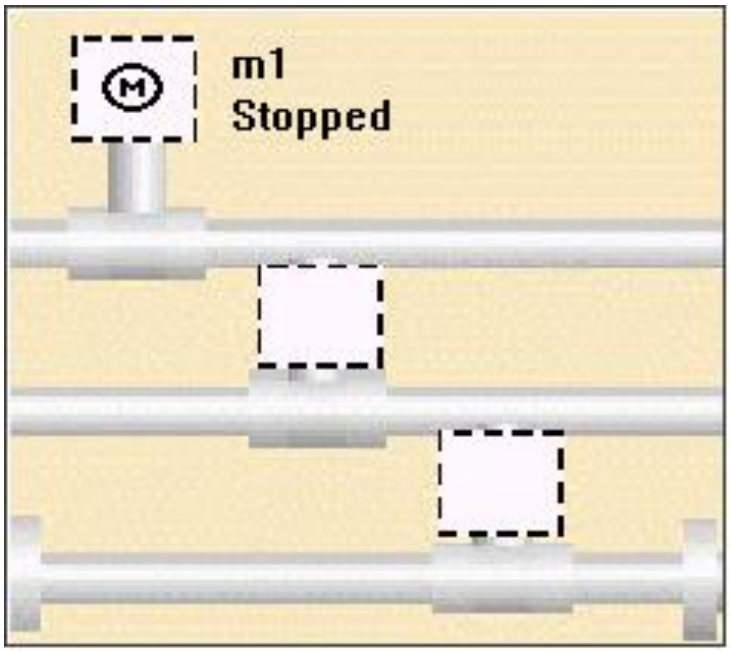
Enter object properties

1. Enter specific PLC addresses for this object.
2. Remaining properties are inherited from the template.



Final object on diagram

1. Place object on diagram.
2. Copy/paste to create more objects.



Try it yourself ...



Exercise: Create templates and template-based objects

Find presentations and exercises on www.7t.dk/igss